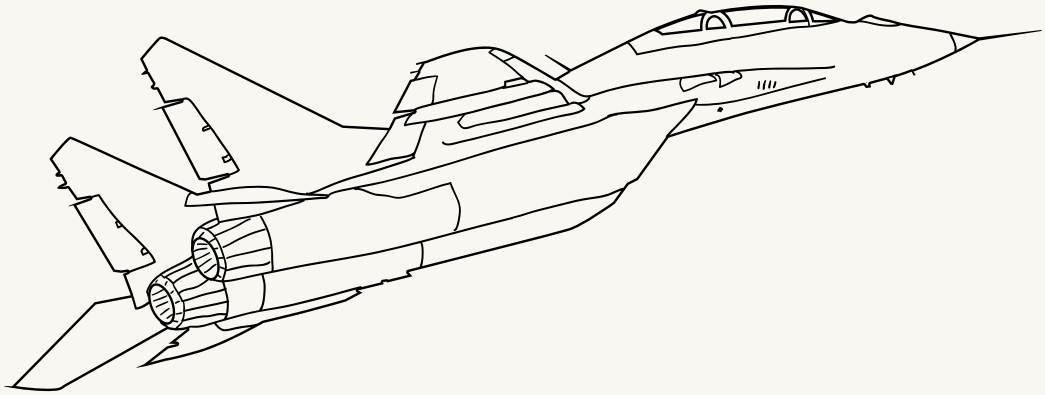


Design and Simulation Portfolio

Vallabh Chandar

vallabhchandar@gmail.com | LinkedIn | Bengaluru, Karnataka

About me

I'm Vallabh, an aerospace and mechanical engineer who loves turning whiteboard doodles into real-world machines. My core focus lies in structural design and analysis for UAVs and airframes, where CAD, FEA, CFD, and physical testing converge to determine what truly performs in the real world.



Over recent years, I've contributed to UAV airframe design, mission payload integration, aeroelastic and aero-thermal simulations. I thrive on challenges where reliability, manufacturability, and practical constraints balance against theoretical elegance.

Beyond engineering, you'll find me practicing my kruthis (a training Carnatic Vocalist) and catching UCL nights with my favourite team, FC Barcelona. I love traveling and pick Mountains > beaches every time ⚽

THAT'S ME ON MY FIRST FLIGHT IN A CESSNA !

Education

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Bachelor's of Engineering in Mechanical Engineering
2019 - 2023

UNIVERSITY OF SHEFFIELD
Master's of Science in Aerospace Engineering
2023 - 2025

What I work with ?

On a day-to-day basis, I live inside CAD and CAE tools: SolidWorks, Fusion 360, Creo, AutoCAD, and ANSYS Mechanical/Fluent for FEA and CFD. My analysis work covers static and dynamic FEA, fatigue and vibration, thermo-structural checks, and load path/reliability assessments for real hardware.

Moving on to some of the cool things I've worked on...

UAS payload integration for post-disaster delivery

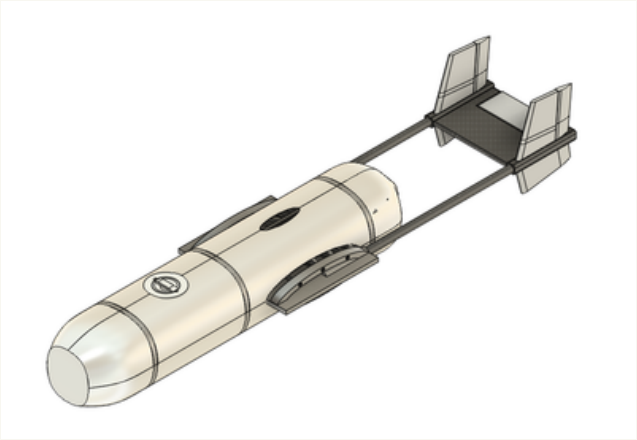
Worked with the National Health Services (NHS) and the University of Sheffield to develop an unmanned gyrocopter to help transport medical supplies between cities in the UK.

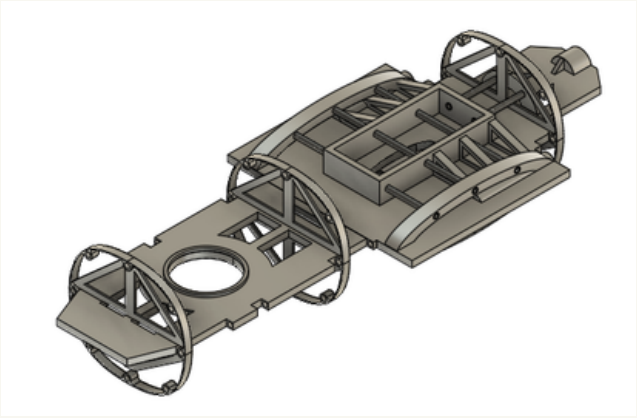
My Role : Structural Engineer

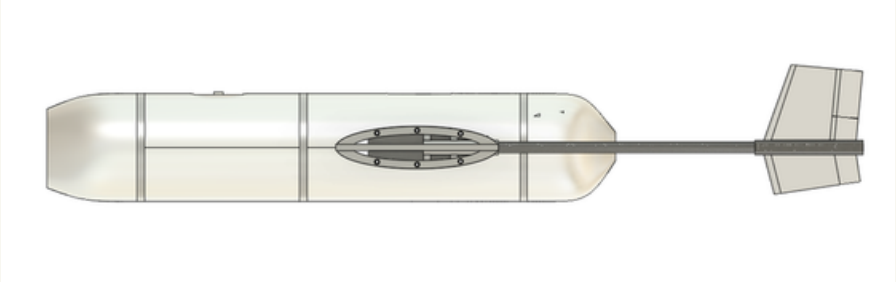
Tools : Fusion360, SolidWorks, Creo, ANSYS Mechanical, MATLAB

What did I do ?

- Designed the primary airframe structure and integration hardware for payload bays, avionics, and subsystem mounts using detailed CAD models, engineering drawings, and Bills of Materials (BOMs).
- Defined structural load paths and performed fatigue analysis at all airframe–subsystem interfaces to ensure vibration tolerance and protect sensitive electronics and medical cargo using ANSYS Mechanical.
- Planned and executed prototype vibration and reliability tests, used test data to validate structural margins, and cleared integration milestones on schedule.
- Documented all design assumptions, interface acceptance criteria, and cross-functional issues, which reduced late-stage rework across aerodynamics, structures, and systems teams.

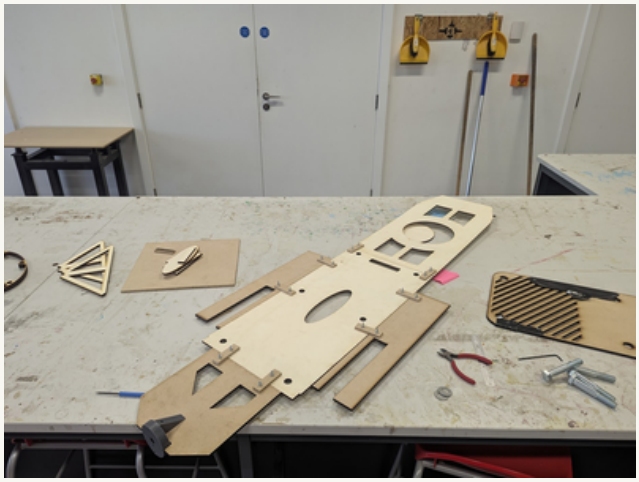


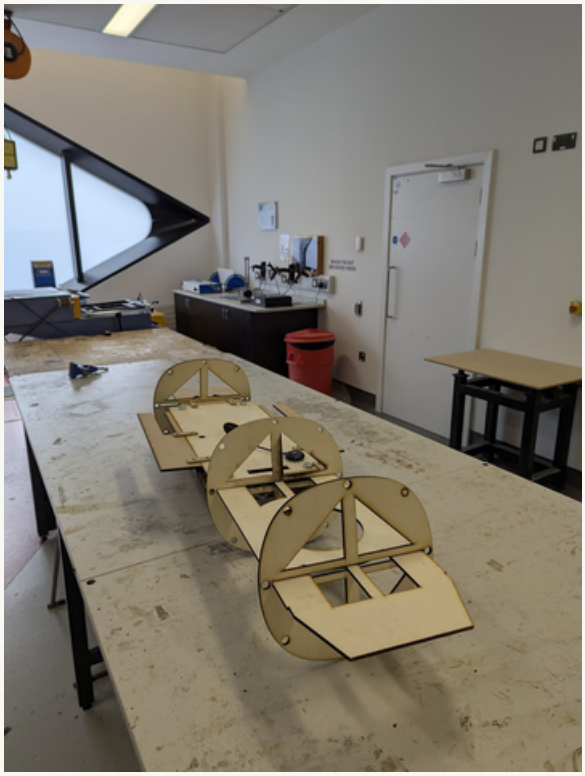




How did I go about it ?

- Clean load paths > fancy shapes – forces should flow, not fight
- Test early, iterate fast – shaker tables beat "it looks strong"
- Document assumptions upfront – saves everyone's weekend





What'd I learn?

FEA + real shaker data = designs that actually work. Good docs = happy teams and using the laser cutter

Unmanned Air System (UAS) for Post-Disaster Payload Delivery

Worked with the School of Mechanical, Aeropsace and Civil Engineering at the University of Sheffield to develop an UAV for Post Disaster Payload Delivery

My Role : Payload and Integration Engineer. Protecting sensitive payloads from getting smashed during rough landings.

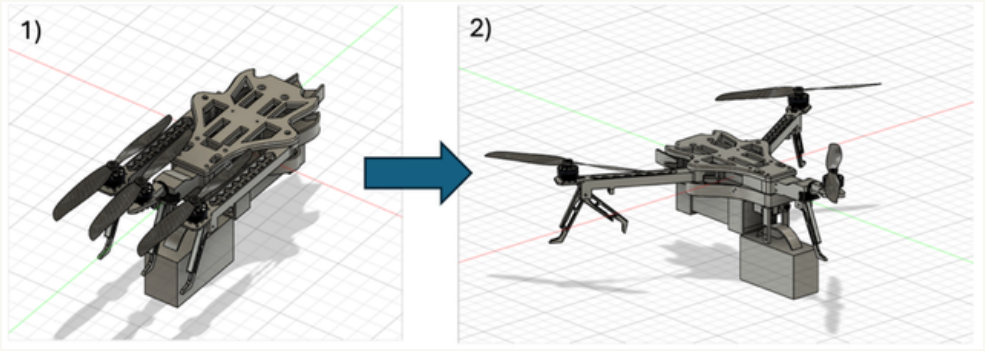
Tools: Fusion360, ANSYS Mechanical, MATLAB

What did I do ?

- Managed payload integration in Fusion – load paths, constraints, clearances so that the gear survived crashes



- Ran reliability structural checks in ANSYS Mechanical on all interfaces/mounts.
- Wrote "certification-style" docs in Python with load cases, safety factors, rules.
- Iterated CAD layouts in SolidWorks to fit mass limits + keep field access easy.

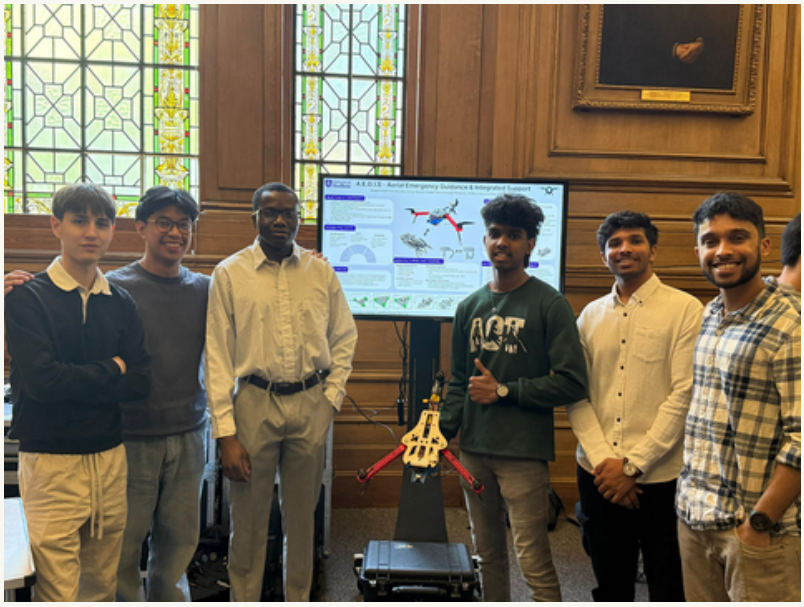


The drone was designed in such a way that the arms could fold making it compact

How did I go about it ?

- Constraints first – design around what's fixed, not wishful
- Safety factors tell a story – pick ones that match real risks
- Field access > pretty CAD – techs fix what's reachable

Finally, one with the crew I worked with in developing the UAV...

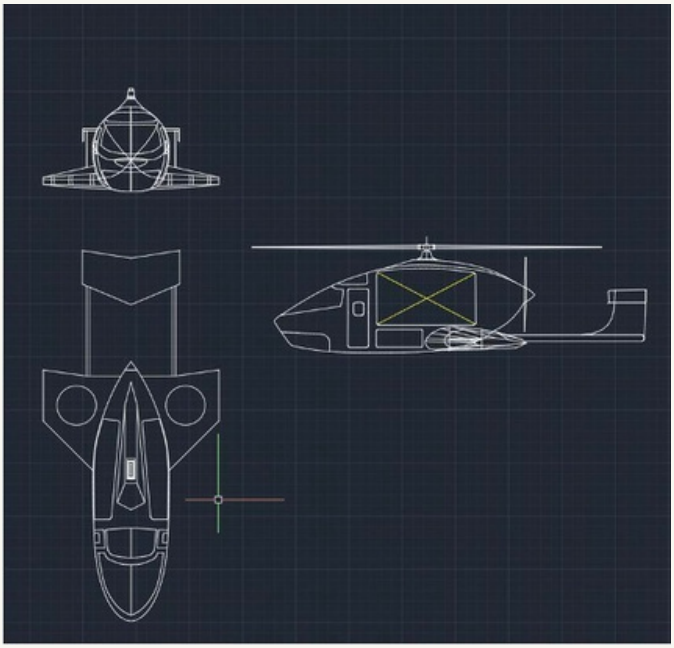


Internship at Aerodyne Aviation Holdings Ltd.

Worked with the Design team to come up with designs for the company's Commercial UAV's that needed production-ready manufacturable drawings

My Role : Design Intern. Turning sketches into shop-floor drawings that suppliers could hit.

Tools : Fusion360, AutoCAD

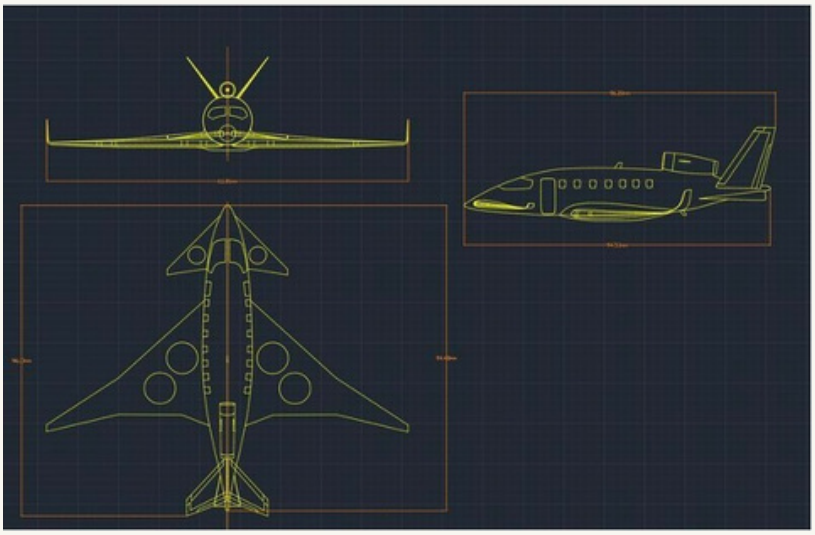


What did I do ?

- Owned subsystem design in SolidWorks/Creo – CAD, hand calcs, ANSYS FEA, GD&T drawings that passed first article inspection.
- Synced tolerances with supplier limits using GD&T standards so parts mated up.
- Wrote test plans with pass/fail criteria in SolidWorks drawings.
- Sequenced builds + defined interfaces with manufacturing/test teams.

How did I go about it ?

- GD&T isn't decoration – it's how parts actually fit
- Supplier calls > perfect drawings – know their limits



What'd I learn ?

Real programs live/die on drawing clarity. Shop floor feedback fixes designs fast.



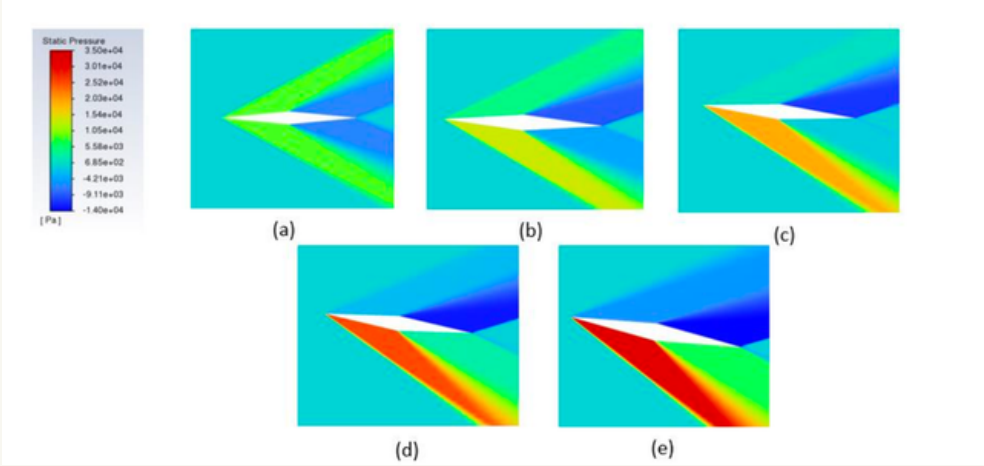
Some of the cool stuff on site at St. Athan’s airfield in Wales, which was where the factory was suppposed to be

Aeroelastic stability of Hypersonic aerofoils

Research into structural stability limits of hypersonic aerofoils under combined dynamic and thermal loads.

My Role : Thesis student involved with FEA modelling and aeroelastic analysis.

Tools : MATLAB, Python, ANSYS Fluent, Solidworks



Pressure distrubution at different angles of attack in a Hypersonic medium

What did I do ?

- Modelled aeroelastic responses in ANSYS Mechanical under combined dynamic + thermal loads to anticipate flutter and forced-response limits.
- Ran convergence + design studies in ANSYS to guide stiffness and geometry choices for subsystem stability.
- Documented every boundary condition, assumption, and validation step in MATLAB for academic review and future reuse.

Teaching, communication and heavy-industry experience

Beyond aerospace labs, I spent time at IIT Bombay developing course content for AE705 (Introduction to Aeronautical Engineering) and AE725 (Air Transportation), turning complex concepts in flight mechanics and air transport systems into structured teaching material. This involved distilling research and technical content into concise notes that improved knowledge transfer for students and research teams.





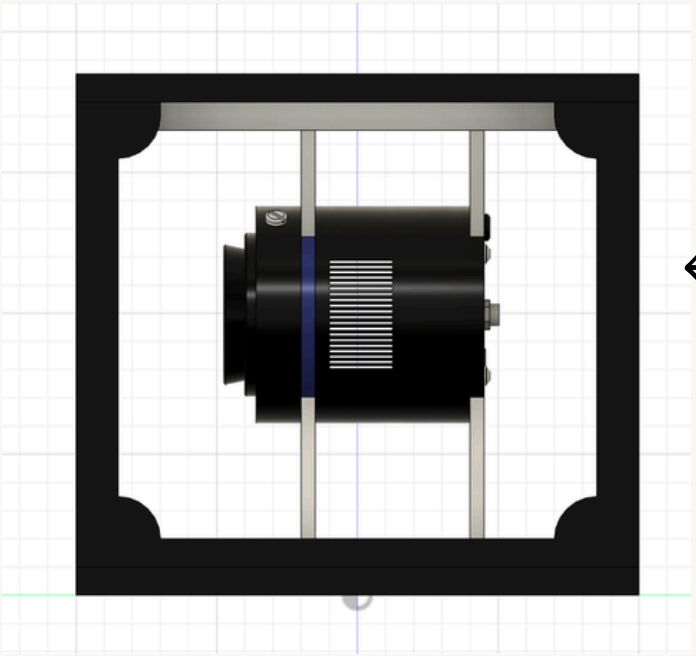
During this internship I got the chance to be a part of the **Design and Engineering of Lighter-Than-Air Systems (DELTAs) Conference 2022** as a part of the team lead by Prof Rajkumar S Pant from the Aerospace Department.



Some projects that I’ve worked on in the past.... so that I didn’t get too rusty

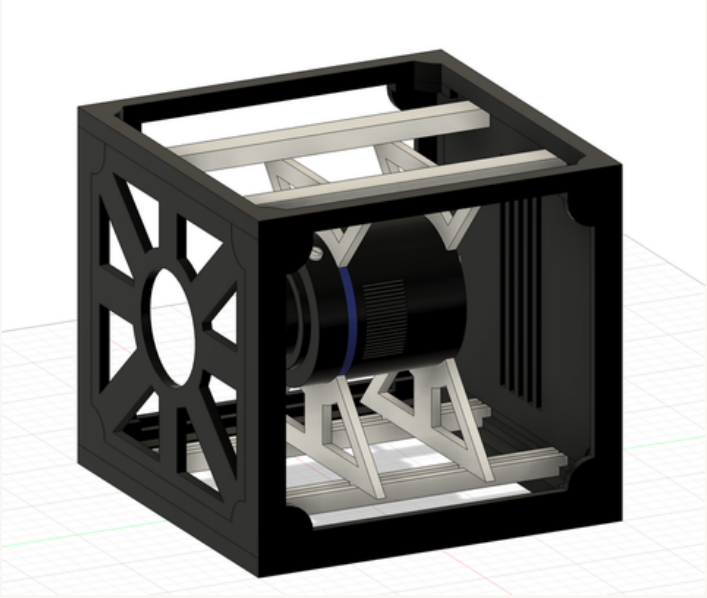
Enclosure for a lens that was to fit into a CUBE SAT

A compact, adjustable enclosure designed for a lens assembly while keeping CubeSat structural constraints in mind. It was developed to support a compact optical setup and still allow flexibility in alignment and adjustment during integration. Didn’t put too much detail into the model as it was made for representation purposes.



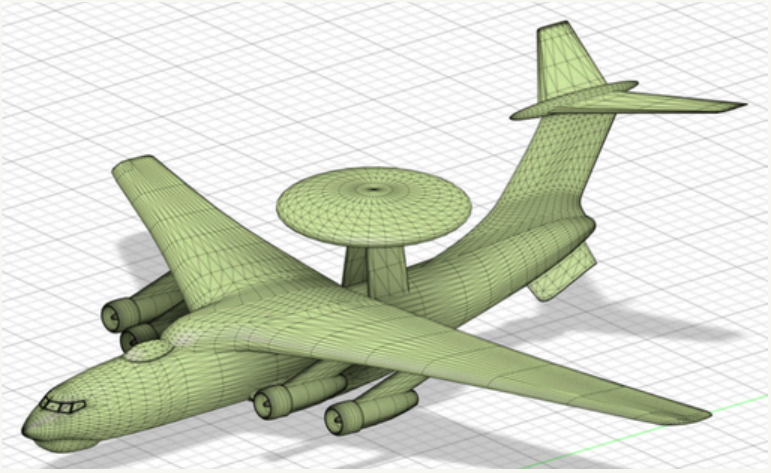
Side view of the enclosure

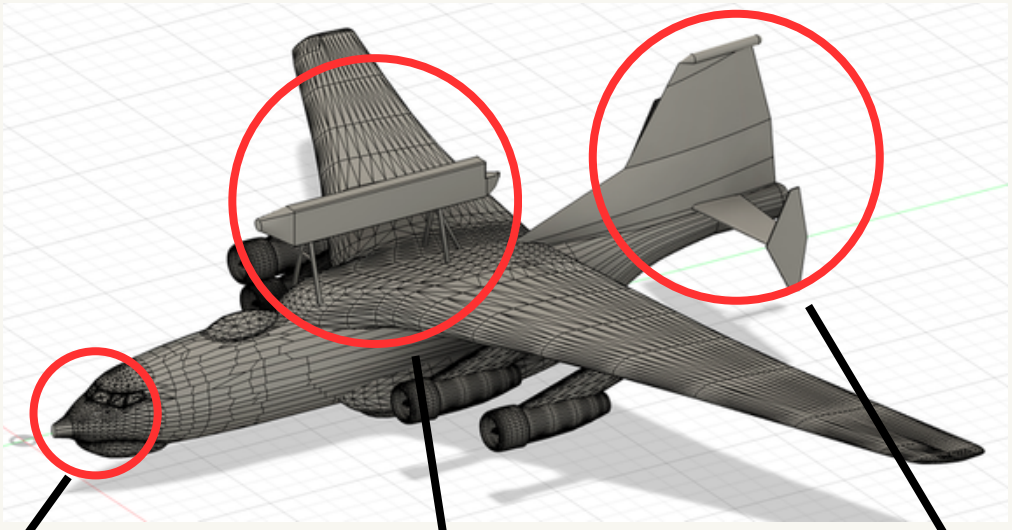
The sliders allow the lens to be readjusted as per the needs

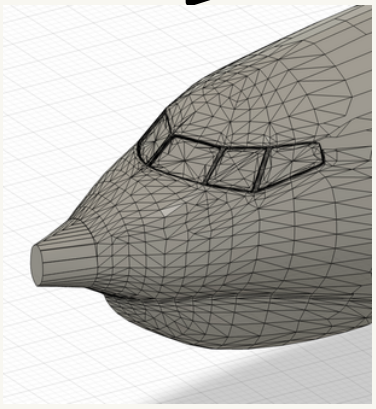


Modifiying a KJ-500 to resemble a KJ-200

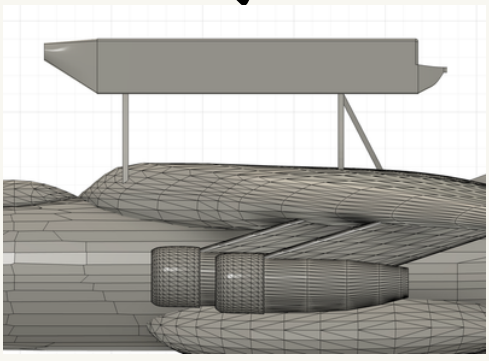
A modified version of the KJ-500 model redesigned to resemble the KJ-200 aircraft. The changes included replacing the radome with a beam radar, adjusting the tail geometry, and modifying the fuselage base from a Y-9 to a Y-8 configuration to better capture the distinct appearance of the KJ-200.



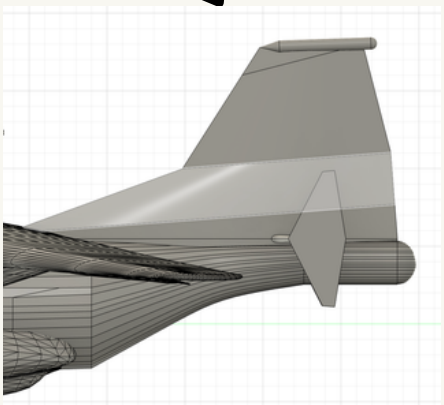




Modified nose cone



Radome to Radar beam



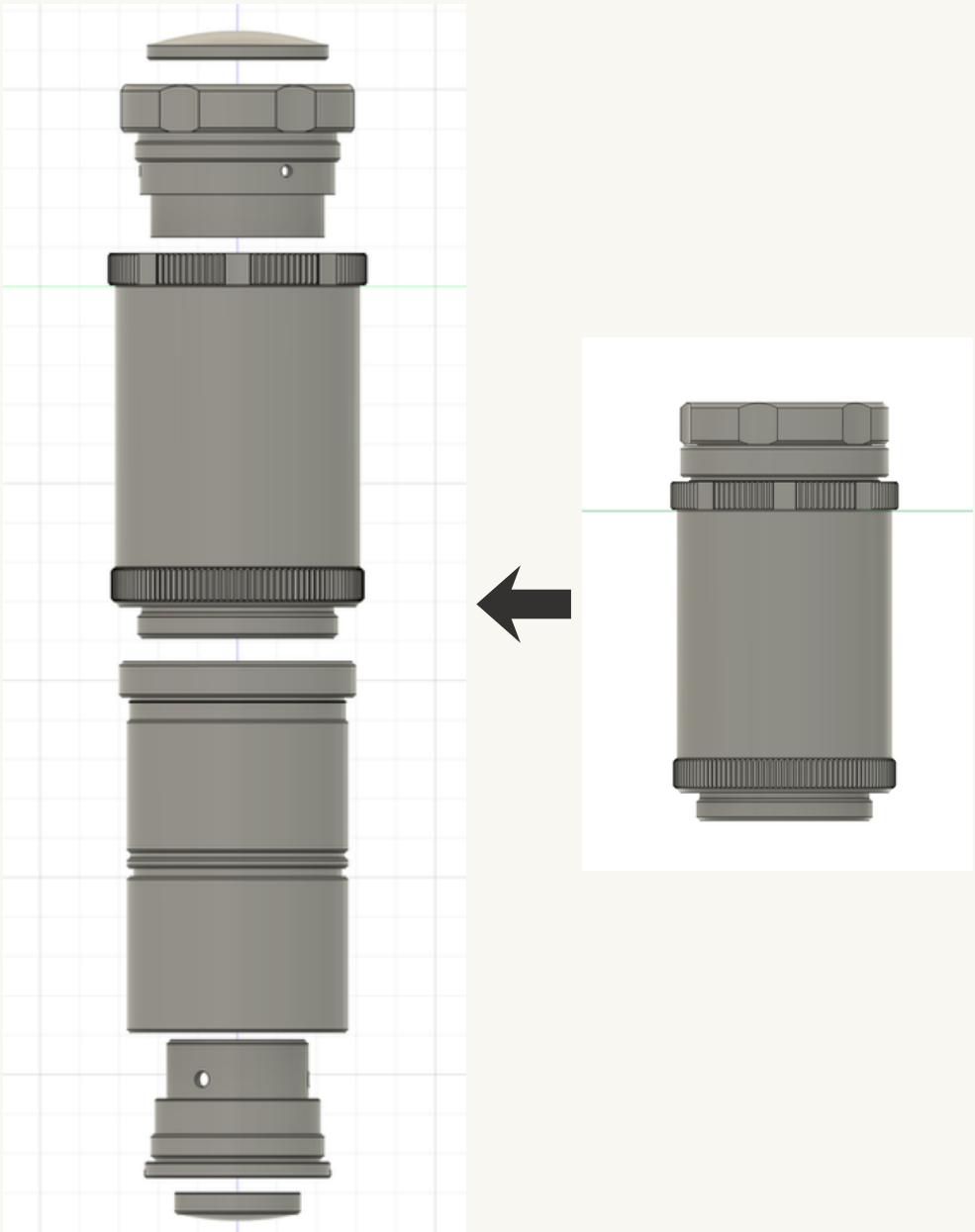
Modified Tail and fuselage base

Kowa Lens constructed from manufacture drawings



A CAD model of a Kowa lens developed for CubeSat applications. The model was created from the manufacturer's specification sheet and refined using GD&T references to ensure dimensional accuracy and proper integration with the surrounding assembly.

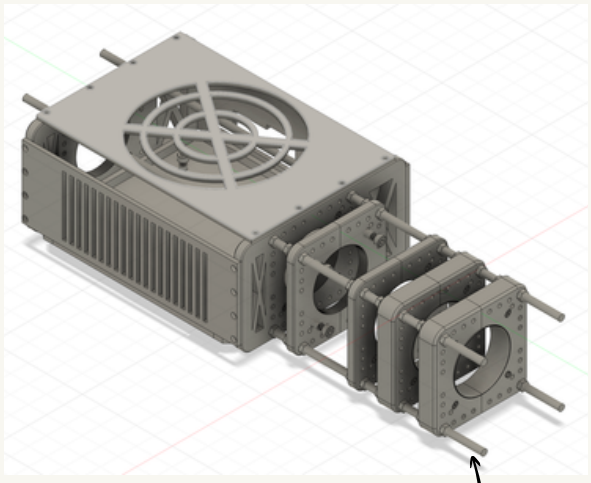
This will be used as a part of the following projects that you'll see next!



An exploded view of the lens's components

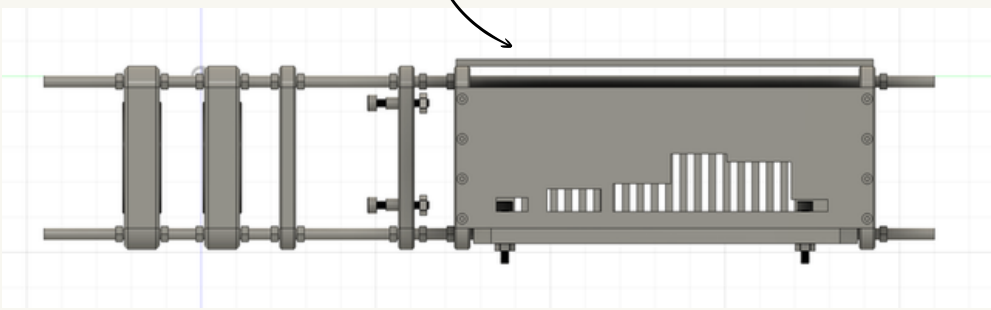
A high-fidelity 3D Engineering Model (EM) of the camera assembly

The first part of this project was to design a structure that could keep the components in place. An internal support structure designed to align and hold the Kowa lens, NVIDIA Jetson, C/CS adapter, and Arducam board. The design was made to allow adjustments when needed, while also enclosing the Jetson in a way that supports heat dissipation and system integration.

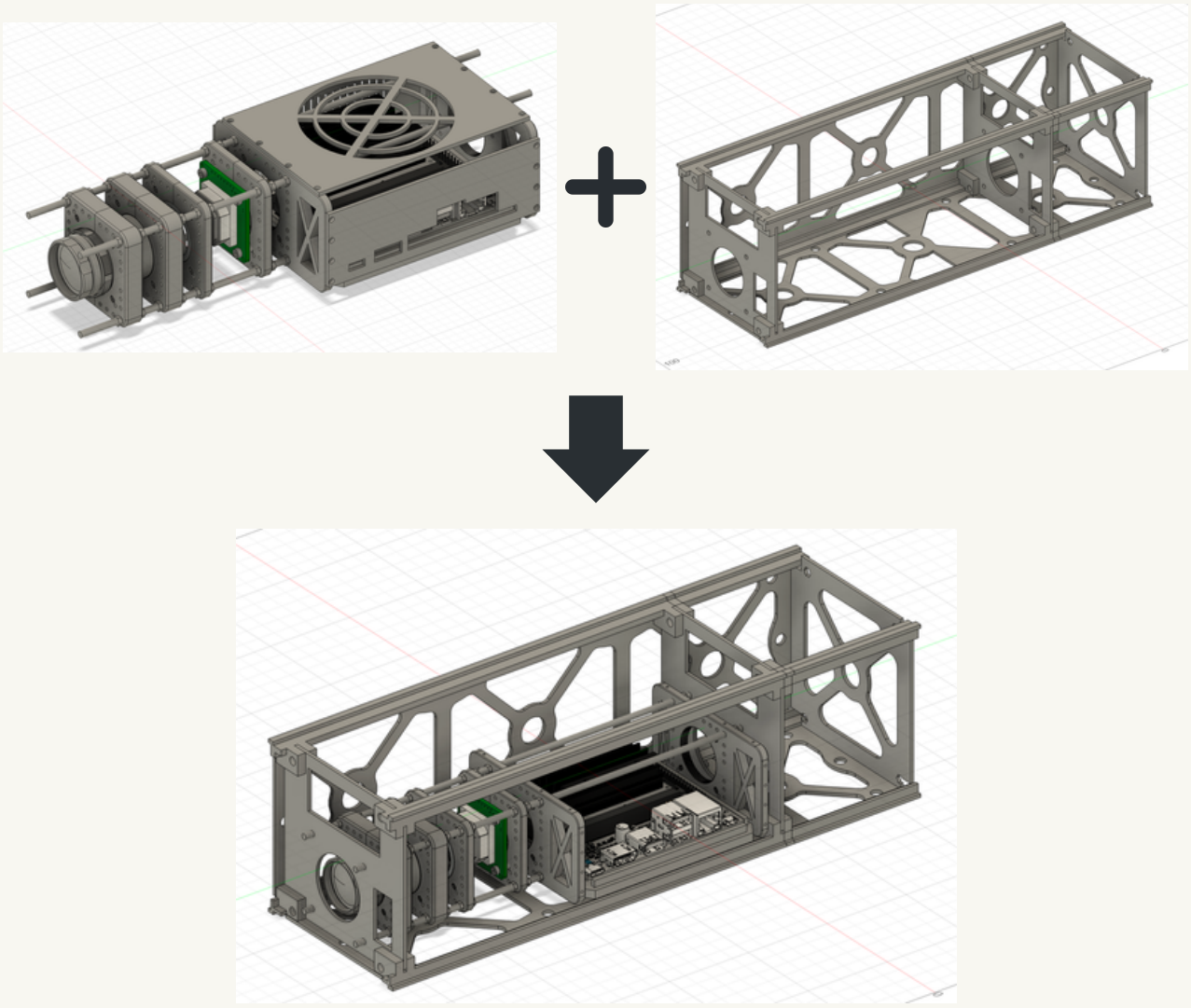


The structure has been divided into two parts:

- The front section which will house the Lens, adapter and the arducam board. This section can be adjusted in terms of the position of the components
- The backend box which houses the Jetson. The main computer on board to process raw sensor data, enhance image resolution, and perform object detection. Cutouts have been made to aid heat dissipation.



This structure was designed to be placed into a 3U frame as illustrated in the designs made below.



The outer 3U structure that houses the lens and associated components. Although only one side of the frame was sourced externally, the geometry was modified as needed to accommodate the inner structure and support the full payload layout.

